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Vision based Hand Gesture and Character Identification using Convolution Neural Network

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***Abstract*— Hand gesture recognition is the interpretation of computer machine with human hand gesture for physically challenged people. This paper proposes a CNN based model for recognizing hand sign language and characters. The image quality is improved by applying the image enhancement technique. This CNN model automatically extracts the discriminative input feature value to produce more accuracy than the traditional model. Fully connected layers are used at the end of the network to perform classification, mapping the learned features to specific ASL character labels. This structured approach ensures the model captures complex patterns within the image data, thereby improving classification performance. To further enhance the accuracy and generalizability of the model, several data augmentation techniques were employed. These techniques include random rotations, flipping, zooming, and shifting of input images. Introducing such variations in the training data, the model becomes more robust to slight differences in image orientation, size, and perspective, reducing the likelihood of overfitting. The proposed technique achieves the classification accuracy of 99.02 percentage on the MNIST dataset.**

Keywords—Hand gesture, Preprocessing, data augmentation, deep learning models.

I. Introduction

In general humans need vocal communication to participate in social activities. Unfortunately, people with speech and/or hearing impairments cannot communicate verbally with others. They use sign language to communicate. Sign Language Recognition (SLR) is used to identify words, numbers, letters, hand gestures, and other signs. Some scientists focus on real-time symbol identification, while others focus on static images. Hand gesture recognition is a vital component of human-computer interaction, allowing users to communicate with machines through intuitive hand movements [1]. This technology plays a significant role in various domains, including sign recognition, virtual reality, gaming, animation and assistive technology for individuals with disabilities. The primary objective of hand gesture recognition is to interpret and classify hand movements accurately, enabling seamless interaction between humans and digital systems. As the demand for touchless interfaces and automation increases, the importance of hand gesture recognition continues to grow. The development of this technology involves advanced techniques in computer vision, image processing, and deep learning to ensure precise recognition and real-time processing [2]. Several techniques are utilized in hand gesture recognition to enhance accuracy and efficiency. The hand gesture-based recognition system was developed by researchers using a variety of technologies, namely extraction with machine feature extraction, machine and deep learning models. Rastgoo, R et.al., [3] proposed a zero shot dynamic hand gesture recognition system, for the feature extraction a transformer-based model with C3D is proposed. Then, for the hand recognition an auto encoder along with the LSTM (long short-term memory) is adopted. Finally, for mapping the semantic space, a bidirectional encoder representation from transformers (BERT) is used. The model is validated using various datasets and achieved the classification rate of74.6 on RKS dataset. Miah, A.S.M et.al.,[4] presented the SLR system technique called GmTC, to translate McSL to text for effective understanding. The feature values are extracted using the graph and deep learning models. First, the graph-based feature values are extracted by considering the super pixel values, then the long and short dependency features are extracted using attention-based technique. The combined features are then passed into the classifiers; it is observed that the SL dataset achieves the higher classification rated compared with other datasets. Jiang, B et. al., [5] proposed SEMG based gesture identification system called NKDFF-CNN. To overcome the drawbacks in the existing kernal convolution operation a narrow kernel model is proposed. The model learns the features from the SEMG between a particular muscle and gesture. Then the feature values are imported to NKDFF-CNN classifier. The proposed technique achieves the classification rate of 85.91% on a multi type hand geature. Al

Mudawi, N et.al., [6] proposed six model techniques like conversion of video to frames, enhancement technique, SSMD (shot multiple box detector), background modelling and finally 1D- convolution neural network (CNN) technique used to classify features. The proposed technique achieves 85.71% on the WLASL datasets. Hassan N er.al., [7] introduced HAR technique (Deep BiLSTM) model to get the unique features from the input images. Then to extract the video feature the deep features CNN and mobileNetV2 is adopted. Finally, the Deep BiLSTM technique is enabled to obtain the optimal classification rate. It achieves the classification rate of 99.20% on the UCF11 dataset. Liang, H et.al.,[8] presented an multi feature learning technique which includes CNN and FAPN (feature aggregation pyramid network), ESA (expansion squeeze aggregation) and task prediction branch. The FAPN and ESA extracts a multiscale feature information at different scales. Miah, A.S.M. et.al., [9] proposed a CNN with multi-stage graph attention and residual connection (GCAR) technique to extract the features. This GCAR technique enhances the features in the skeleton points, this effectively extract the entire body movement features effectively and feeds into the two-stream information classifier. The proposed technique achieves 90.31% on the WLASR datasets. This same

II. DEEP LEARNING ALGORITHM

Deep learning algorithms have fundamentally transformed the field of computer vision, offering remarkable accuracy in tasks such as image classification, object detection, and gesture recognition. Among these, Convolutional Neural Networks (CNNs) are particularly effective for gesture recognition due to their ability to process and extract hierarchical features from visual data, making them the optimal choice for applications involving image recognition. CNN network has several layers to extract the features from the image database. CNN applies the different kernals and filters to the image to learn the pattern of images like edges, textures and corners [10]. To enhance the precision of CNN network, the false positive values are reduced. The precision also depends on the labels of the particular dataset. Increasing the number of the filters in the CNN classifier improves the performance.

When the input image crosses the layers in the CNN network, it automatically learns the complex features from the input image to identify the unique feature values. To reduce the number of inputs features the pooling layers are placed next to the convolution layer to reduce the dimensionality. This is very important to reduce the computational complexity and reduces the overfitting problem[11].At the end the fully connected layer is placed, this helps to produce the final output by taking the all the features which is extracted by convolution layer.This layered approach, where each successive layer captures increasingly complex features, makes CNNs particularly effective for tasks like gesture recognition, where fine-grained distinctions between hand shapes and movements need to be made. The basic workflow diagram of the proposed work is presented in Fig.1. The methodology adopted in this project is structured into several key stages: dataset collection and organization, preprocessing, image enhancement, segmentation, data augmentation, deep learning model design, and training/testing procedures.

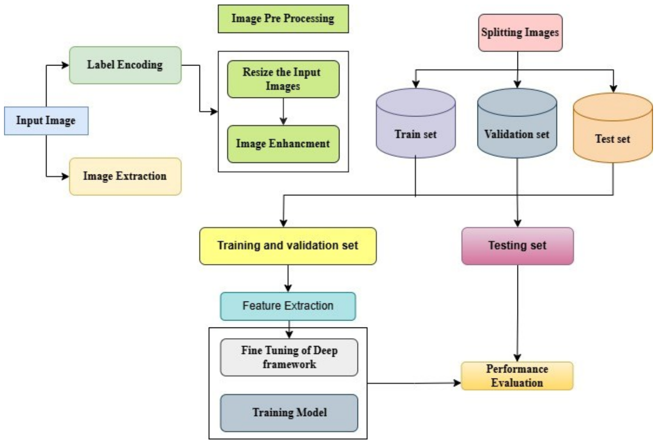

Fig.1 Workflow Diagram

The dataset forms the foundation of any machine learning or deep learning project. For this study, we utilized a combination of publicly available datasets images to ensure diversity in hand gestures, lighting conditions, and backgrounds. A well-organized and diverse dataset is essential for training a robust model capable of handling real-world variations. We leveraged existing datasets such as the Sign Language MNIST dataset, which contains grayscale images of American Sign Language (ASL) characters. Additionally, other datasets like ASL Alphabet and Hand Gesture Recognition Dataset were incorporated to enrich the variety of gestures. These datasets provided a solid baseline for training and validation. American Sign Language dataset includes 2400 images, 1800 for training and 600 for testing images.

In next step the image enhancement techniques were applied to improve image clarity and emphasize critical features of hand gestures, contributing to better model accuracy. Contrast adjustment histogram equalization improves contrast, making gesture features more prominent. Brightness normalization techniques are used to balance across images to mitigate lighting inconsistencies. To reduce the noise, median filtering and Gaussian filtering were applied to remove unwanted noise. Finally, the sharpening technique is used to sharpen the images the edge details to aid in better feature extraction [12]. These techniques collectively ensured the visibility of key features in the dataset.

To increase the size of the dataset and to enhance the model performance, several augmentation techniques were applied like rotation cropping, flipping, zooming and adding noise to the input image [13]. The sample augmented image is shown in Fig.2

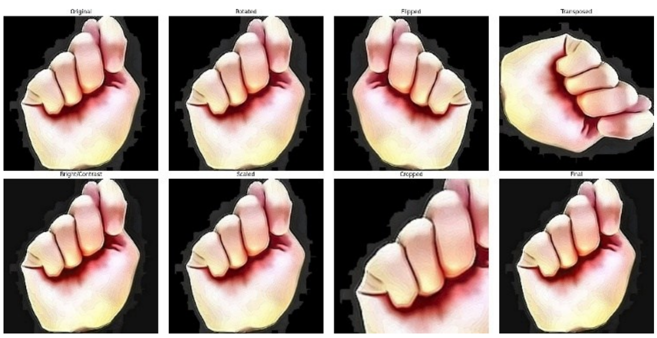

Fig.2 Data Augmentation

To improve the quality of the image various improvement techniques like sharpening, denising, brightness, contrast adjustment are applied. The sample images are shown in Fig.3

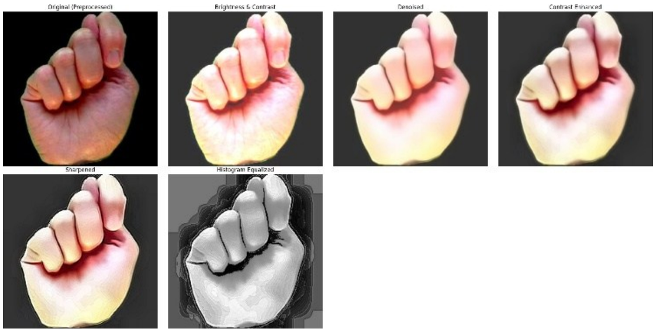

Fig.3 Image Enhancement

III. Methodology

The implementation of the proposed work starts with the preprocessing of the images. In this preprocessing stage first images are formatted and resized uniformly. Then the image enhancement technique is applied to reduce the noise, color correction, contrast adjustment to enhance the quality of the input images to get more accurate results. Next, data augmentation is applied to increase the size of the dataset. In the final stage, the images are then trained with the proposed CNN technique. The proposed technique processes the images in multiple layers and learns the input feature pattern for different hand gestures. The each layer for the CNN is important to get more accurate results. The proposed system CNN architecture is presented in Fig.4.

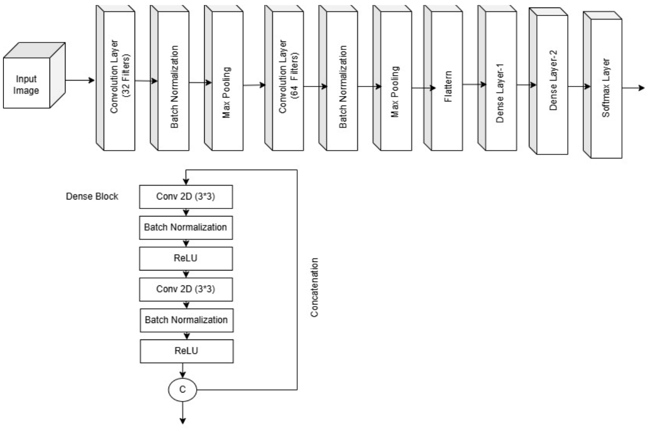

Fig.4 Block Diagram of the proposed Technique

A neural network may consist of a hardware device and computer code that mimics how neurons function in the human brain. The imaging process is not a good fit for artificial neural networks. CNN employs a system that is similar to a multilayer view-point and was created with fewer processing requirements in mind. Eliminating restrictions and boosting image processing power results in a system that is far more efficient and makes it simpler to train data for both image processing and language communication. A much-improved CNN model has been proposed with nine layers in our CNN model. The CNN model has fourteen stages, including hidden layers, which provide us the best possible outcome for hand sign detection.

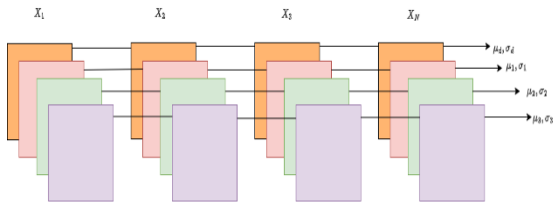

Fig.5 Batch Normalization

In order to create unvarying dimensions, the proposed method has taken a wide range of images as input and transformed them all into a single, constant size of 128*128*3. Typically, we develop a convolutional kernel by applying thirty-two 2*2 convolutional filters at the input layer, each supported by three channel tensors. Then the ReLU activation function is applied to make a decision on the output [14]. ReLU will produce the output if the values are positive, else zero is transferred to the output. Next, batch normalization is introduced as the next layer. The batch normalization technique normalizes the layer inputs by re-centering and rescaling, which not only produces neural networks more quickly but also adds stability as shown in Fig.5. The pooling process then entails accumulating the features that fall inside the filter's coverage area after applying a 2D filter to each feature map. The dimensions of the output layer are obtained with $b_h \times b_w \times b_c$

$$\frac{b_h - f_s + 1}{S} \times \frac{b_w - f + 1}{S \times b_c}$$

Where b_h - Feature map height, b_w - Feature map width, b_c - No of features in the channel map, f filter size, S length of the stride. Many convolutional and pooling layers are stacked one after the other in a typical CNN model design. The pooling layers help reduce the dimensions of the feature maps, which is their primary advantage. The pooling layer model is presented in Fig.6. Consequently, it prevents overfitting and reduces the computational cost. In CNNs, the pooling layer lowers the model's parameter count. This layer eliminates noise or small fluctuations in the input data while capturing the key elements of the image. Consequently, it gives the model more resilience to size, rotation, and blockage changes.

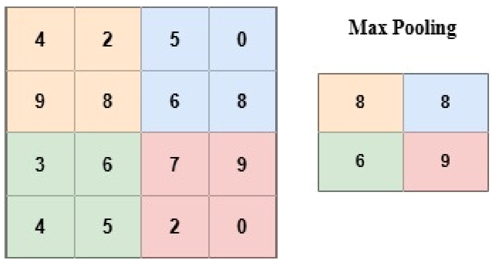

Fig.6 Max Pooling

In next step Prior to flattening, once more 64 filter convolutional, batch normalization, and max-pooling layers are employed. Then, the two dense layers are used with the final two levels in the second dense layer and 512 concealed layers in the first. Finally, softmax provides greater precision than other activation functions, we employed it in the last layer [15].

IV. RESULTS AND DISCUSSION

This section explains the experimental findings and results for the proposed techniques. To implement the proposed method, the Keras API and TensorFlow 2.x are used to train the proposed system. The selected framework has several features and is specifically made for convolutional neural networks (CNNs), the most widely used architecture in the gesture detection challenge. The model is integrated with the adam optimizer to change the learning rate dynamically. The optimizer dynamically adjusts the learning rate for different parameters in the CNN model, which helps the model converge more quickly and avoid local minima. The learning rate is originally set to 0.001 in order to balance learning speed and stability. Learning rate schedules can be used to dynamically alter the learning rate based on the performance attained during training. The problem has several classes (each gesture is a class), hence a categorical cross entropy model loss function is utilized. The variation of the values between the original and the predicted probability distribution across all classes is calculated using categorical cross entropy. The model is trained using five epochs. The batch size used in this model is 32. The training of the model is assessed by plotting the training and validation loss and accuracy over the epochs. These plots reveal information regarding overfitting and underfitting as well as the model's learning process. It is observed from the model accuracy graph how the training data works on the validation dataset to get the model more accurate value.

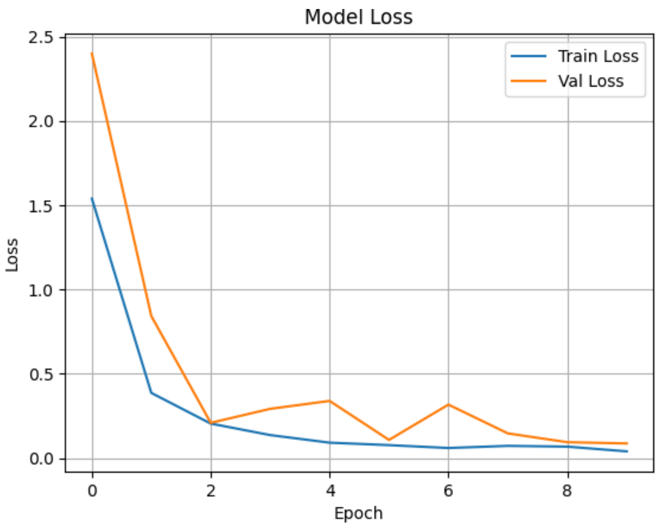

Fig.7 Model Accuracy Graph

Depending on the size of the training dataset and the amount of augmented training data, the model's accuracy varied upto and 98%. This range of accuracy suggests that the model can accurately detect hand movements in a variety of test samples, demonstrating its high predictability. The model accuracy and loss values are shown in Fig.7 and 8.

A dedicated prediction script was designed to evaluate the performance of the trained CNN model in classifying previously unseen hand gesture images. This script loads a single image, processes it to meet the input requirements of the model, and then classifies it using the trained model. The prediction image was pre-processed to a uniform input size of 96×96 pixels and normalize the values from 0 to 1.

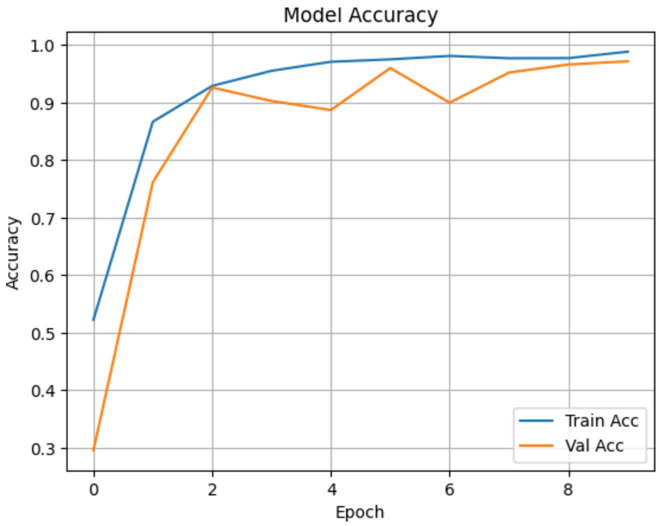

Fig.8 Model Loss Graph

This ensures consistency throughout the training and validation process. The model then processes the image to give a probability vector showing the probability of each class. The most probable class is selected as the ultimate prediction. An instance of Image Data- Generator is employed to get the mapping from numerical class indices to their corresponding labels as in the class organization of the training directory. This ensures that the displayed label is the same as that of the original dataset's class organization shown in Fig.9

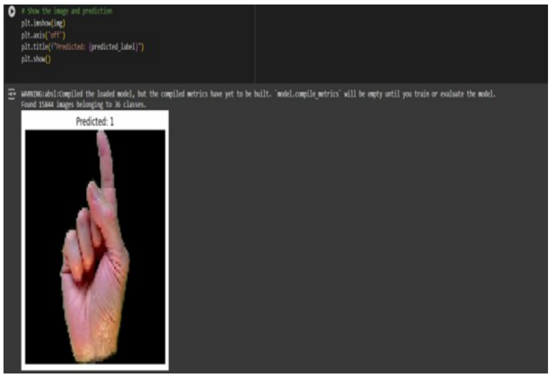

Fig. 9 Model Predictions

V. CONCLUSION AND FUTURE WORK

In this study, the CNN model is utilized to classify the hand gesture sign model. To avoid the overfitting problem the data augmentation technique is used. The proposed CNN model reduces the process of detection, feature extraction and segmentation. This CNN model automatically learns the discriminative features from the images to give higher accuracy. The proposed custom CNN model shows the accuracy variation up to 98 percent on various datasets. Then the prediction model is developed based on the proposed technique to make an accurate prediction of different gestures. To improve the accuracy further the deeper and transfer learning model will be adopted for human gesture recognition.

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